

Endometriosis

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FIBOG

Students should know:

- What is Endometriosis?
- Incidence & common sites
- Causes and Pathogenesis
- Pathology
- Clinical presentation, Infertility!!!
- Investigations
- Differential diagnosis
- Treatment
- Prognosis
- Prevention

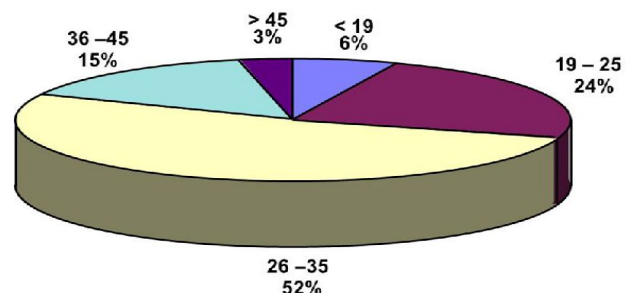
Definition:

Endometriosis is a common gynecological condition which is defined as endometrial tissue lying outside the endometrial cavity. Endometriotic tissue responding to hormonal changes and therefore undergoes cyclical bleeding during menstrual periods which with time causing Chronic Inflammatory process where it seeded eg. Peritoneal cavity.

Incidence:

- Most common benign gynecological condition
- 10-15% of all women population
- 8-10% of reproductive age
- 30-40 % of endometriosis cases have Infertility
- 20-50% of infertility cases have Endometriosis
- 40-50% of patients with chronic pelvic pain have endometriosis

Age preference: —————→



Common sites:

We can divide the seeding sites of endometrial tissue extrauterine into:

1) Usually seen in peritoneal cavity and **Predominantly** within the pelvis and much **Commonly** on the uterosacral ligaments because it's the most dependent place of pelvic cavity. The second most common site which is mainly appear as cystic lesion is ovary. *Other pelvic* organs include ureter, rectovaginal septum, rectum and rarely bladder

2) Other extrapelvic (uncommon) sites include umbilicus, abdominal scars, nasal passages, pleural cavity, pericardium.

Aetiology and pathogenesis

The aetiology of endometriosis is unknown, although there have been many theories, and in each case more than one may be blamed.

I) Sampson's implantation theory (John Albertson Sampson (1873–1946)): It postulates retrograde menstrual regurgitation of viable endometrial glands and tissue within the menstrual fluid and subsequent implantation on the peritoneal surface.

II) Coelomic epithelium transformation: Meyer's 'coelomic metaplasia' theory (Dr. Robert Meyer (1864–1947)) describes the de-differentiation of peritoneal cells lining the Müllerian duct back to their primitive origin which then transform into endometrial cells. This may be due to hormonal stimuli or inflammatory irritation.

III) Genetic, immunological and environmental factors: It has been suggested that genetic and immunological factors may alter the susceptibility of a woman and allow her to develop endometriosis. The incidence in first-degree relatives of patients seen to be common and racial differences, with increased incidence among oriental women more than Afro-Caribbean origin. Numerous studies implicate that exposure to environmental toxins like 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) may play a role in development of endometriosis.

IV) Vascular and lymphatic spread: Vascular and lymphatic embolization to distant sites has been demonstrated and explains the rare findings of endometriosis in sites outside the peritoneal cavity.

Pathological appearance:

1) Superficial 'powder-burn' or 'gunshot' lesions: These lesions appear like black, dark-brown or bluish puckered lesions, containing old haemorrhage surrounded by a variable extent of fibrosis, white plaques or scarring of peritoneal discoloration.

2) Deep infiltration: Deeply infiltrating endometriotic nodules extend > 5 mm beneath the peritoneum and may involve the USL ligaments, vagina, bowel, bladder or ureters.

3)Endometriomas: These cystic lesion contain thick, like chocolate, fluid / densely adherent to the peritoneum of the ovarian fossa (fibrosis) may involve the tubes and bowel (Chocolate cyst).

Clinical presentation:

Endometriosis may be asymptomatic but the classical clinical features are severe cyclical non-colicky pelvic pain restricted to around the time of menstruation, sometimes associated with heavy menstrual loss. Symptoms may begin a few days before menses starts until the end of menses. There is a lack of correlation between extent of disease and the intensity of symptoms.

Symptoms include: —————→

- Pain (Dys-, symptoms), **Dysmenorrhea**,
- Dyspareunia**, **Dyschesia**, **Dysuria**
- Nausea, vomiting, easy fatigability.
- Inflammatory bowel syndrome like features (colicky abdominal pain, bloating)
- Hematuria
- Infertility
- Chronic pelvic pain

Site	Symptoms
Female reproductive tract	Dysmenorrhoea Lower abdominal and pelvic pain Dyspareunia Rupture/torsion endometrioma Low back pain Infertility
Urinary tract	Cyclical haematuria/dysuria Ureteric obstruction
Gastrointestinal tract	Dyschezia (pain on defecation) Cyclical rectal bleeding Obstruction
Surgical scars/umbilicus	Cyclical pain and bleeding
Lung	Cyclical haemoptysis Haemopneumothorax

Mechanisms of infertility in Endometriosis:

It's multifactorial:

- Anatomic distortion and adhesions in cystic lesions
- Ovarian function
- Tubal function
- Sperm function —————→
- Coital function
- Early pregnancy failure

Physical examination:

It is better to be performed during menstruation. The examination is best be done in bimanual abdominal, using speculum and rectovaginally(PR, PV).

Ovarian function	Luteolysis caused by prostaglandins F2 Oocyte maturation defects Endocrinopathies Luteinized unruptured follicle syndrome Altered prolactin release Anovulation
Tubal function	Impaired fimbrial oocyte pick-up Altered tubal mobility
Coital function	Deep dyspareunia – reduced coital frequency
Sperm function	Antibodies causing inactivation Macrophage phagocytosis of spermatozoa
Early pregnancy failure	Prostaglandin induced Immune reaction Luteal phase deficiency

The usual findings are:

- Thickening or nodularity of the uterosacral ligament
- Tenderness in the pouch of Douglas
- Adnexal mass
- Fixed retroverted uterus
- Visible lesions in vagina or cervix or other areas

Investigations:

1) Ultrasonography: whether trans- abdominal or vaginal give us a perfect information regarding gross pathologies like chocolate cysts or fluid collection in cul de sac and the site of involvement. Its of lesser use in smaller lesions like superficial and deep infiltration.

2) MRI: MRI can detect lesions >5 mm in size, particularly in deep tissues, for example rectovaginal septum. This can allow careful presurgical planning in difficult cases.

3) Laparoscopy + Histology: It is the gold standard diagnostic test, especially in doubtful cases of chronic pelvic pain, we can differentiate between the causes or presence of more than one cause.

Advantage: Direct vision of the lesion (Diagnostic), and we can add therapeutic methods also like cyst excision, diathermy, LASER ablation.

Disadvantage: Subjective experience which may take us to the entity of personal error.

4) Markers: For example CA 125, CA 19-9, Interleukin-6 (IL-6) are nonspecific measures in the diagnosis of endometriosis.

Differential Diagnosis: →

Treatment:

To understand the treatment strategies, we should answer the following questions:

- What are you treating (disease, symptoms or both)?
- Why are you treating?

The possible reasons for treatment are:

- Improve natural fertility
- Enhance chances of success in ART

Gynecologic

Pelvic inflammatory disease
Tuboovarian abscess
Salpingitis
Endometritis
Hemorrhagic ovarian cyst
Ovarian torsion
Primary dysmenorrhea
Degenerating leiomyoma
Ectopic pregnancy
Other pregnancy complications

Nongynecologic

Interstitial cystitis
Chronic urinary tract infection
Renal calculi
Inflammatory bowel disease
Irritable bowel syndrome
Diverticulitis
Mesenteric lymphadenitis
Musculoskeletal disorders

- Pain relief as an alternative to surgery
- Pain relief while awaiting surgery
- Adjunct to surgery
- Prophylaxis against disease occurrence
- Symptoms recurrence

There are some realities that the patient and her relatives should know about the nature of the pathology and its treatment:

- Patients often difficult to treat and it is impossible to guarantee complete cure.
- Therapies designed for long-term strategies and tailored for each patient situation.
- Coexisting additional diseases such as irritable bowel/ constipation should also be treated to improve overall success rates.

Medical therapy:

*Analgesics: NSAIDs are the group of drugs used for relieving different pain symptoms of endometriosis, eg. Naproxin, Ibuprofen,....

Important factors affecting treatment choices:

- Woman's age
- Nature of symptoms and severity of disease
- Desire of child-bearing
- Priorities and attitude
- Other infertility factors

Codeine and opiates are exacerbating IBS like features, so better to be avoided.

*Combined Oral Contraceptive (COC): Also called HRT (Hormonal Replacement Therapy), used for diagnostic and therapeutic purposes for a period of 6 months initially. Mechanism of action:

COC → suppress ovarian function → causing amenorrhea → reduces pain and other symptoms

If there is symptomatic relief with the continuous use of COC, so the therapy should be continued for up to several years or even longer.

Drug choices include:

(*) **Progestogens**: works by antagonizing estrogen effect on endometrium → endometrial atrophy → amenorrhea.

Example of drugs, Ceralette, MPA (Medroxy Progesteron Acetate), LNG-IUS (levonorgestrel intrauterine systems), SPRM (Selective Progesteron Receptor Modulators). These drugs are the best choice for long term chronic cases use particularly after surgical treatment.

(*) **Danazol/gestrinone**: These drugs are Synthetic Androgen → suppression of mid-cycle LH surge → chronic anovulatory state.

They are less commonly used because of side effects, such as androgen effects, for example weight

gain, greasy skin and acne over long term (>six months), alterations in lipid profiles or liver function.

(*)Gonadotrophin-releasing hormone agonists: They work by Decreasing circulating level of oestrogen → pseudomenopause.

Example of drugs Lupron depot, Trelstar, Zoladex. Side effects include symptoms seen at the menopause, in particular hot flushes and night sweats. Despite these side effects, the drugs are well tolerated and they have become established agents in the treatment of endometriosis. Long-term use (>six months) can lead to drug-induced osteoporosis, limiting its use.

Surgical therapy:

Conservative surgery:

This choice of surgical treatment is suitable for woman who want to complete her family. The technique used here is:

Laparoscopic surgery with techniques such as diathermy, laser vaporization, adhesolysis or excision has become the standard for the surgical management of endometriosis.

Due to the Recurrence rate of pathologies by >30%, concurrent long-term medical therapy is usually useful.

Definitive surgery:

Indications:

- Woman whose family is complete
- Severe symptoms
- Progressive disease

Hysterectomy and bilateral salpingo- oophorectomy is done which is usually curative. The removal of the ovaries is essential in achieving long-term symptom relief.

Continue combined HRT for up to six months following surgery, particularly when active disease was found to be present at the time of laparotomy, to prevent activation of any residual disease.

Infertility treatment

Important points:

- Mild Endometriosis → get benefit from conservative laparoscopy
- Moderate to Severe → need surgical correction like cyst removal or restore tubal function

Assisted Reproductive Techniques (ART) include:

- Controlled Ovarian Hyperstimulation eg. Clomiphene
- IUI (IntraUterine Insemination)
- IVF (In Vitro Fertilization)

Other surgical Treatments

For treatment of severe intractable pain:

- Presacral Neurectomy
- Laparoscopic Uterine Nerve Ablation (LUNA)

Prognosis:

- Depending on severity, location, fertility desire
- 75% of mild case are fertile while 30-40% may gain fertility in proper management in moderate to severe cases
- Reaching early diagnosis and treating the case has important role in improving pain and fertility

Prevention:

- Early child-bearing
- Oral contraceptive pills
- Healthy life and exercise

Adenomyosis

Short notes:

- Presence of endometrial tissue deep in myometrium
- 10-15%
- Most seen in late 30s or early 40s of age women
- Present as severe secondary dysmenorrhea and menorrhagia
- Tender bulky “boggy” uterus
- Diagnosis ,by U/S but better by MRI
- Medical and conservative surgery to gain amenorrhea poorly developed
- Hysterectomy is the definitive treatment